

A CLIQUE-CORE COUNTEREXAMPLE TO A DIRECTED CONNECTIVITY CONJECTURE

ABSTRACT. A bidirected clique placed between sources and sinks disproves the proposed formula for the maximum size of a simple digraph with bounded local arc-connectivity. The counterexamples exist for every $m \geq 5$ and every $m + 7 \leq n \leq 3m - 3$. The smallest member of this family has twelve vertices, fifty-seven arcs, and maximum local arc-connectivity four.

A digraph is *simple* if it has no loops or parallel arcs; opposite arcs are allowed. Write $\lambda_D(u, v)$ for the maximum number of pairwise arc-disjoint directed u - v paths, and let $\ell_m^{\text{dir}}(n)$ be the maximum number of arcs in a simple n -vertex digraph satisfying $\lambda_D(u, v) \leq m - 1$ for all distinct u, v . Conjecture 3.1 of [1] proposes, for $n \geq m \geq 2$,

$$(1) \quad \ell_m^{\text{dir}}(n) = \max \left\{ m(n-1), \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \right\}.$$

The following elementary construction shows that this is false.

Theorem 1. *For all integers $n \geq m \geq 2$,*

$$\ell_m^{\text{dir}}(n) \geq B_m(n) := \left\lfloor \frac{(n+m-3)^2}{4} \right\rfloor + 2(m-1).$$

In particular, (1) fails whenever $m \geq 5$ and $m + 7 \leq n \leq 3m - 3$.

Proof. Put $r = n - m$, $a = \lceil r/2 \rceil$ and $b = \lfloor r/2 \rfloor$. Take disjoint sets S, C, T of sizes a, m, b , respectively, and label $C = \{c_1, \dots, c_m\}$. Use exactly the following arcs:

$$c_i \rightarrow c_j \ (i \neq j), \quad S \rightarrow \{c_1, \dots, c_{m-1}\}, \quad \{c_1, \dots, c_{m-2}\} \rightarrow T, \quad S \rightarrow T.$$

An arrow between sets means that every corresponding arc is present. Thus C induces a bidirected clique, S consists of sources, and T consists of sinks.

Here are cuts certifying every possibly nonzero local connectivity. The source side X of a cut contains the starting vertex and not the terminal vertex; its capacity is the number of arcs from X to its complement.

Ordered pair	Source side X	Capacity
$c_i, c_j \ (i \neq j)$	$\{c_i\} \cup T$	$m - 1$
s, c_j	$\{s\} \cup T$	$m - 1$
c_i, t	$C \cup (T \setminus \{t\})$	$m - 2$
s, t	$\{s\} \cup C \cup (T \setminus \{t\})$	$m - 1$

Every other ordered pair is unreachable. Every family of arc-disjoint paths uses distinct arcs of a separating cut, so all local arc-connectivities are at most $m - 1$. Equality holds between distinct core vertices: the direct arc and the $m - 2$ two-step paths through the other core vertices are arc-disjoint.

Since $a(b+1) = \lfloor (r+1)^2/4 \rfloor$, the arc count is

$$\begin{aligned} |A(D)| &= m(m-1) + (m-1)a + (m-2)b + ab \\ &= m(m-1) + (m-2)r + \left\lfloor \frac{(r+1)^2}{4} \right\rfloor = B_m(n). \end{aligned}$$

For comparison with the two conjectured branches, direct subtraction gives

$$(2) \quad B_m(n) - m(n-1) = \left\lfloor \frac{(r+1)^2}{4} \right\rfloor - 2r,$$

$$(3) \quad B_m(n) - \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor = m-1 - \left\lfloor \frac{r}{2} \right\rfloor.$$

The first difference is positive for $r \geq 7$, and the second is positive for $r \leq 2m-3$. Both conditions hold in the stated range. $\square$

The twelve-vertex example. Take $m = 5$, $|S| = 4$ and $|T| = 3$. The four arc classes above contain 20, 16, 9 and 12 arcs, respectively. Hence

$$|A(D)| = 57, \quad \lambda_D^{\max} = 4, \quad \max\{5 \cdot 11, \lfloor 15^2/4 \rfloor\} = 56.$$

This is a counterexample for *arc-disjoint* paths, not merely for internally vertex-disjoint paths. No minimality claim is made for its order.

Corollary 2. *If, for a fixed $m \geq 5$, the identity*

$$\ell_m^{\text{dir}}(n) = \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor$$

holds for every $n \geq n_0(m)$, then $n_0(m) \geq 3m-2$.

Proof. At $n = 3m-3$, equation (3) gives an excess of one over the proposed quadratic value. $\square$

The construction does not refute eventual exactness for fixed m : its quadratic-branch excess in (3) is zero at $n = 3m-2, 3m-1$ and negative thereafter. Nor does it refute the $m = 3, 4$ special cases. Its conclusion is a new lower bound, not an exact determination of $\ell_m^{\text{dir}}(n)$.

REFERENCES

- [1] Louis Vandenbruwaene, *Maximising Edge Density Subject to Connectivity Constraints: Erdős Problem 915, from Proof to Search*, master's thesis manuscript, KU Leuven, academic year 2026–2027, Conjecture 3.1 (uploaded version `main (11).pdf`).